# Review of Submission 9D

First Proof Second Batch

Reviewer 1

June 4–8, 2026

## 1 Review

**Correctness.** This paper has some of the correct ingredients for this problem, but fails to articulate them correctly or connect them into a proof. Some of the statements are wrong, and some flat-out nonsensical, like evaluating a coinvariant algebra at a superspace.

**Novelty.** It is difficult to speak to the novelty of a partial result, especially when no combinatorial interpretation is given. Mentioning the superspace coinvariant ring and ordered set partitions shows a base level of insight into the problem. Using Cauchy's decomposition on $\mathrm{Sym}(V \otimes E)$ could be an interesting approach, if it worked.

**Attribution.** This paper does not cite any sources besides one incorrect inline citation, which makes following and verifying the line of reasoning even more difficult.

## 2 Recommendation

This paper should be rejected. The paper itself admits that it is only a partial attempt at a solution and, accordingly, the paper fails to create a line of logic, justify steps, or reference any results, all while having errors. Plus, it does not arrive at a combinatorial interpretation.

## 3 Submission with inline comments

**Disclaimer:** The following document presents a partial attempt at solving the given problem. The automated verification pipeline did not confirm a complete solution. The argument is transcribed here faithfully, maintaining its current state, including any unverified claims or logical gaps.

### Problem Statement

Let $m$ and $n$ be positive integers and let $R_n^{(m)}$ be the coinvariant algebra with $m$ sets of $n$ variables.

The Hilbert series of an $m$-multigraded algebra $A$ is

$$\mathrm{Hilb}(A; q_1, \ldots, q_m) = \sum_{i_1,\ldots,i_m \geq 0} \dim(A_{(i_1,\ldots,i_m)}) q_1^{i_1} \cdots q_m^{i_m},$$

where $A_{(i_1,\ldots,i_m)}$ denotes a multigraded component of the algebra, with grading variables $q_1, \ldots, q_m$.

Let $s_\lambda(q_1, \ldots, q_m)$ denote a Schur polynomial. For $R_n^{(m)}$, it is known that its Hilbert series may be expressed by

$$\mathrm{Hilb}(R_n^{(m)}; q_1, \ldots, q_m) = \sum_\lambda c_\lambda(n) s_\lambda(q_1, \ldots, q_m),$$

for some nonnegative universal series coefficients $c_\lambda(n)$ which do not depend on $m$.

For any $n$, give a combinatorial interpretation of the coefficients $c_\lambda(n)$ when $\lambda$ is a hook shape partition.

## Proposed Solution

**Problem Statement Reframing:** Let $m$ and $n$ be positive integers and let $R_n^{(m)}$ be the coinvariant algebra with $m$ sets of $n$ variables. Give a combinatorial interpretation of the universal nonnegative series coefficients $c_{(k,1^d)}(n)$ in its $m$-multigraded Hilbert series when $\lambda$ is a hook shape partition.

*Partial Proof.* The following sequence of steps provides a partial argument outlining a proposed interpretation:

1. Let $\mathbb{C}[X] = \mathrm{Sym}(V \otimes E)$ be the polynomial ring representing $m$ sets of $n$ variables, where $V$ is the $(n-1)$-dimensional standard representation of $S_n$ and $E = \mathbb{C}^m$. The multi-graded coinvariant algebra is $R_n^{(m)} = \mathrm{Sym}(V \otimes E)/I$, where the ideal $I$ is generated by the positive-degree multi-symmetric invariants.

   > **Reviewer 1**
   >
   > $V$ should be $n$-dimensional, not $(n-1)$-dimensional.

2. By Cauchy's formula, $\mathrm{Sym}(V \otimes E)$ decomposes as a $GL(V) \times GL(E)$ module into $\bigoplus_\lambda S^\lambda(V) \otimes S^\lambda(E)$. Because the quotient ideal $I$ is $S_n$-invariant, the $GL(E)$ character of $R_n^{(m)}$ decomposes identically as $\sum_\lambda c_\lambda(n) s_\lambda(E)$. The coefficients $c_\lambda(n)$ are universal non-negative integers that do not depend on $m$ (provided $m \geq \ell(\lambda)$).

   > **Reviewer 1**
   >
   > The notation $s_\lambda(E)$ probably refers to a Schur functor, but warrants more explanation.

3. To strictly isolate the hook shape partitions $\lambda = (k, 1^d)$, we evaluate the $GL_m$ character at the superspace $E = \mathbb{C}^{1|1}$ (one commuting variable $q$, one anti-commuting variable $z$). In this space, the Schur functor $S^\lambda(\mathbb{C}^{1|1})$ evaluates to zero for all non-hook shapes.

> **Reviewer 1**
>
> What does it mean to evaluate a $GL_m$ character at a superspace? They might be trying to get a plethysm here. Even if so, there would still be some missing characters. What is the justification for the second sentence?

4. Evaluating $R_n^{(m)}$ at $E = \mathbb{C}^{1|1}$ maps the generators of the multi-symmetric invariant ideal $I$ (the polarizations of the power sums $p_r$) directly to $p_r(x)$ and their fermionic derivations $\sum x_i^{r-1}\theta_i$. This quotient is mathematically identical to the super-coinvariant algebra

$$SR_n = \mathbb{C}[x_1, \ldots, x_n] \otimes \Lambda(\theta_1, \ldots, \theta_n) / \left\langle p_r(x), \sum x_i^{r-1}\theta_i \right\rangle .$$

> **Reviewer 1**
>
> Again, it is unclear what it means to evaluate $R_n^{(m)}$ at $E$.

5. Consequently, the universal bosonic hook coefficient $c_{(k,1^d)}(n)$ is exactly the multiplicity of the super-Schur module $S^{(k,1^d)}(\mathbb{C}^{1|1})$ in the super-coinvariant algebra $SR_n$.

> **Reviewer 1**
>
> What is a super-Schur module?

6. For any hook of size $\geq 1$, the irreducible $GL(1|1)$ module $S^{(k,1^d)}(\mathbb{C}^{1|1})$ has dimension 2, spanned by a highest-weight vector and its lowering derivation. It is a canonical combinatorial theorem (e.g., Haglund–Rhoades–Shimozono) that the total dimension of $SR_n$ is exactly $\mathrm{OSP}(n)$, the number of Ordered Set Partitions of $[n]$.

> **Reviewer 1**
>
> Haglund–Rhoades–Shimozono have multiple papers together, and none of them have this result.

7. **Combinatorial Interpretation:** The universal coefficient $c_{(k,1^d)}(n)$ exactly enumerates the number of Ordered Set Partitions of $[n]$ that act as highest-weight "sources" (elements annihilated by the fermionic derivation operator in the OSP combinatorial basis of $SR_n$) possessing exactly a polynomial degree of $k$ and an exterior degree of $d$.

> **Reviewer 1**
>
> This is not a combinatorial interpretation, as it relies on a vague algebraic statement. Plus, I am not aware of an OSP combinatorial basis of $SR_n$.

*(Note on the $m = 1$ collapse: For $d = 0$, the exterior degree is zero, which strictly filters the domain to Ordered Set Partitions possessing exactly $n$ blocks. An Ordered Set Partition of $[n]$ into exactly $n$ blocks is bijectively a permutation in $S_n$. Thus, for $d = 0$, the combinatorial interpretation perfectly collapses to counting permutations by their major index $k$, yielding the exact Mahonian distribution $c_{(k)}(n)$ without violating cardinality bounds).* □

# Review of Submission 9D

First Proof Second Batch
Reviewer 2
June 4–8, 2026

## 1 Review

**Summary.** This submission included a disclaimer at the top that it does not solve the problem. It does not, however, point out where it went wrong. It lists out thoughts towards a potential strategy, but many of the steps do not make sense. The final proposed solution is in terms of a basis which may or may not exist (the "OSP combinatorial basis of $SR_n$", which is not defined and has not appeared explicitly in the literature). Its proposed "combinatorial" interpretation of the coefficients $\{c_{(a,1^b)}(n)\}$ is the number of these basis elements annihilated by some operator.

**Correctness.** This paper is incomplete, and certain steps are clearly incorrect.

**Novelty.** The (incomplete) solution "feels" more like an algebraic interpretation instead of a combinatorial one – see final inline comment for what I mean by this.

**Attributions.** This submission includes no bibliography, and the only citation (an in-line citation) references the wrong paper.

**Exposition.** This submission is written as a list of ideas, and it does not define several of the notations used.

## 2 Recommendation

I would recommend this submission be rejected.

## 3 Submission with inline comments

**Disclaimer:** The following document presents a partial attempt at solving the given problem. The automated verification pipeline did not confirm a complete solution. The argument is transcribed here faithfully, maintaining its current state, including any unverified claims or logical gaps.

## Problem Statement

Let $m$ and $n$ be positive integers and let $R_n^{(m)}$ be the coinvariant algebra with $m$ sets of $n$ variables.

The Hilbert series of an $m$-multigraded algebra $A$ is

$$\mathrm{Hilb}(A; q_1, \ldots, q_m) = \sum_{i_1,\ldots,i_m \geq 0} \dim(A_{(i_1,\ldots,i_m)}) q_1^{i_1} \cdots q_m^{i_m},$$

where $A_{(i_1,\ldots,i_m)}$ denotes a multigraded component of the algebra, with grading variables $q_1, \ldots, q_m$.

Let $s_\lambda(q_1, \ldots, q_m)$ denote a Schur polynomial. For $R_n^{(m)}$, it is known that its Hilbert series may be expressed by

$$\mathrm{Hilb}(R_n^{(m)}; q_1, \ldots, q_m) = \sum_\lambda c_\lambda(n) s_\lambda(q_1, \ldots, q_m),$$

for some nonnegative universal series coefficients $c_\lambda(n)$ which do not depend on $m$.

For any $n$, give a combinatorial interpretation of the coefficients $c_\lambda(n)$ when $\lambda$ is a hook shape partition.

## Proposed Solution

**Problem Statement Reframing:** Let $m$ and $n$ be positive integers and let $R_n^{(m)}$ be the coinvariant algebra with $m$ sets of $n$ variables. Give a combinatorial interpretation of the universal nonnegative series coefficients $c_{(k,1^d)}(n)$ in its $m$-multigraded Hilbert series when $\lambda$ is a hook shape partition.

*Partial Proof.* The following sequence of steps provides a partial argument outlining a proposed interpretation:

1. Let $\mathbb{C}[X] = \mathrm{Sym}(V \otimes E)$ be the polynomial ring representing $m$ sets of $n$ variables, where $V$ is the $(n-1)$-dimensional standard representation of $S_n$

> **Reviewer 2**
>
> Do you mean that $V$ is $n$-dimensional? As written, $X$ would have to refer to $m$ sets of $(n-1)$ variables, rather than $n$.

   and $E = \mathbb{C}^m$. The multi-graded coinvariant algebra is $R_n^{(m)} = \mathrm{Sym}(V \otimes E)/I$, where the ideal $I$ is generated by the positive-degree multi-symmetric invariants.

2. By Cauchy's formula, $\mathrm{Sym}(V \otimes E)$ decomposes as a $GL(V) \times GL(E)$ module into $\bigoplus_\lambda S^\lambda(V) \otimes S^\lambda(E)$. Because the quotient ideal $I$ is $S_n$-invariant, the $GL(E)$ character of $R_n^{(m)}$ decomposes identically as $\sum_\lambda c_\lambda(n) s_\lambda(E)$. The coefficients $c_\lambda(n)$ are universal non-negative integers that do not depend on $m$ (provided $m \geq \ell(\lambda)$).

3. To strictly isolate the hook shape partitions $\lambda = (k, 1^d)$, we evaluate the $GL_m$ character at the superspace $E = \mathbb{C}^{1|1}$

> **Reviewer 2**
>
> Should say what you mean by evaluating "the" $GL_m$ character *at* $E$ – I would guess you are meaning to apply the Schur functor $S^\lambda$?

(one commuting variable $q$, one anti-commuting variable $z$). In this space, the Schur functor $S^\lambda(\mathbb{C}^{1|1})$ evaluates to zero for all non-hook shapes.

> **Reviewer 2**
>
> This is not obvious, so it needs a reference or a proof!

4. Evaluating $R_n^{(m)}$ at $E = \mathbb{C}^{1|1}$

> **Reviewer 2**
>
> What does it mean to "evaluate" a ring "at" $E$?

maps the generators of the multi-symmetric invariant ideal $I$ (the polarizations of the power sums $p_r$) directly to $p_r(x)$ and their fermionic derivations $\sum x_i^{r-1}\theta_i$. This quotient is mathematically identical

> **Reviewer 2**
>
> Unimportant, but "mathematically identical" is unusual phrasing...

to the super-coinvariant algebra

$$SR_n = \mathbb{C}[x_1, \dots, x_n] \otimes \Lambda(\theta_1, \dots, \theta_n) / \left\langle p_r(x), \sum x_i^{r-1}\theta_i \right\rangle .$$

5. Consequently, the universal bosonic hook coefficient $c_{(k,1^d)}(n)$ is exactly the multiplicity of the super-Schur module $S^{(k,1^d)}(\mathbb{C}^{1|1})$ in the super-coinvariant algebra $SR_n$.

6. For any hook of size $\geq 1$, the irreducible $GL(1|1)$ module $S^{(k,1^d)}(\mathbb{C}^{1|1})$ has dimension 2, spanned by a highest-weight vector and its lowering derivation. It is a canonical combinatorial theorem (e.g., Haglund–Rhoades–Shimozono) that the total dimension of $SR_n$ is exactly $\mathrm{OSP}(n)$, the number of Ordered Set Partitions of $[n]$.

> **Reviewer 2**
>
> This is the wrong reference – this is proved in the Rhoades–Wilson Hilbert series paper instead. Haglund–Rhoades–Shimozono prove something different (but related).

7. **Combinatorial Interpretation:** The universal coefficient $c_{(k,1^d)}(n)$ exactly enumerates the number of Ordered Set Partitions of $[n]$ that act as highest-weight "sources" (elements annihilated by the fermionic derivation operator in the OSP combinatorial basis of $SR_n$)

> **Reviewer 2**
>
> I don't think there's an agreed-upon canonical "OSP combinatorial basis" for $SR_n$. The only basis I've seen isn't presented as indexed by OSPs, but rather by monomials fitting under certain "$J$-staircase." It's of course possible the AI has a specific bijection in mind to translate, but this would have to be stated and explained. Also, why the number of annhilated basis elements matches the multiplicity of the desired module in $SR_n$ needs to be proved too.
>
> Finally, even if the gaps can be proved, this doesn't "feel" like a combinatorial answer – I would say that one would need to do one more step where they combinatorially characterize the OSP whose associated basis elements get annhilated.

possessing exactly a polynomial degree of $k$ and an exterior degree of $d$.

*(Note on the $m = 1$ collapse: For $\mathrm{d} = 0$, the exterior degree is zero, which strictly filters the domain to Ordered Set Partitions possessing exactly $n$ blocks. An Ordered Set Partition of $[n]$ into exactly $n$ blocks is bijectively a permutation in $S_n$. Thus, for $\mathrm{d} = 0$, the combinatorial interpretation perfectly collapses to counting permutations by their major index $k$, yielding the exact Mahonian distribution $c_{(k)}(n)$ without violating cardinality bounds).* $\square$

# Review of Submission 9D

First Proof Second Batch
Reviewer 3
June 4–8, 2026

## 1 Review

The answer does not give a combinatorial formula, but only suggests a strategy that might lead to one. However, the steps of this strategy are not explained clearly and do not seem to be well connected.

The final remark on the case $d = 0$ is correct, but no reference is given.

**Correctness.** The ideas suggested are basically correct, but the statements are usually inaccurate, the ideas are not well connected and references are missing.

## 2 Recommendation

I recommend that the submission be rejected.

## 3 Submission with inline comments

**Disclaimer:** The following document presents a partial attempt at solving the given problem. The automated verification pipeline did not confirm a complete solution. The argument is transcribed here faithfully, maintaining its current state, including any unverified claims or logical gaps.

### Problem Statement

Let $m$ and $n$ be positive integers and let $R_n^{(m)}$ be the coinvariant algebra with $m$ sets of $n$ variables.

The Hilbert series of an $m$-multigraded algebra $A$ is

$$\mathrm{Hilb}(A; q_1, \ldots, q_m) = \sum_{i_1,\ldots,i_m \geq 0} \dim(A_{(i_1,\ldots,i_m)}) q_1^{i_1} \cdots q_m^{i_m},$$

where $A_{(i_1,\ldots,i_m)}$ denotes a multigraded component of the algebra, with grading variables $q_1, \ldots, q_m$.

Let $s_\lambda(q_1, \ldots, q_m)$ denote a Schur polynomial. For $R_n^{(m)}$, it is known that its Hilbert series may be expressed by

$$\mathrm{Hilb}(R_n^{(m)}; q_1, \ldots, q_m) = \sum_{\lambda} c_\lambda(n) s_\lambda(q_1, \ldots, q_m),$$

for some nonnegative universal series coefficients $c_\lambda(n)$ which do not depend on $m$.

For any $n$, give a combinatorial interpretation of the coefficients $c_\lambda(n)$ when $\lambda$ is a hook shape partition.

## Proposed Solution

**Problem Statement Reframing:** Let $m$ and $n$ be positive integers and let $R_n^{(m)}$ be the coinvariant algebra with $m$ sets of $n$ variables. Give a combinatorial interpretation of the universal nonnegative series coefficients $c_{(k,1^d)}(n)$ in its $m$-multigraded Hilbert series when $\lambda$ is a hook shape partition.

*Partial Proof.* The following sequence of steps provides a partial argument outlining a proposed interpretation:

1. Let $\mathbb{C}[X] = \mathrm{Sym}(V \otimes E)$ be the polynomial ring representing $m$ sets of $n$ variables, where $V$ is the $(n-1)$-dimensional standard representation of $S_n$ and $E = \mathbb{C}^m$. The multigraded coinvariant algebra is $R_n^{(m)} = \mathrm{Sym}(V \otimes E)/I$, where the ideal $I$ is generated by the positive-degree multi-symmetric invariants.

2. By Cauchy's formula, $\mathrm{Sym}(V \otimes E)$ decomposes as a $GL(V) \times GL(E)$ module into $\bigoplus_\lambda S^\lambda(V) \otimes S^\lambda(E)$. Because the quotient ideal $I$ is $S_n$-invariant, the $GL(E)$ character of $R_n^{(m)}$ decomposes identically as $\sum_\lambda c_\lambda(n) s_\lambda(E)$. The coefficients $c_\lambda(n)$ are universal non-negative integers that do not depend on $m$ (provided $m \geq \ell(\lambda)$).

3. To strictly isolate the hook shape partitions $\lambda = (k, 1^d)$, we evaluate the $GL_m$ character at the superspace $E = \mathbb{C}^{1|1}$ (one commuting variable $q$, one anti-commuting variable $z$). In this space, the Schur functor $S^\lambda(\mathbb{C}^{1|1})$ evaluates to zero for all non-hook shapes.

4. Evaluating $R_n^{(m)}$ at $E = \mathbb{C}^{1|1}$ maps the generators of the multi-symmetric invariant ideal $I$ (the polarizations of the power sums $p_r$) directly to $p_r(x)$ and their fermionic derivations $\sum x_i^{r-1}\theta_i$. This quotient is mathematically identical to the super-coinvariant algebra

$$SR_n = \mathbb{C}[x_1, \ldots, x_n] \otimes \Lambda(\theta_1, \ldots, \theta_n) / \left\langle p_r(x), \sum x_i^{r-1}\theta_i \right\rangle.$$

5. Consequently, the universal bosonic hook coefficient $c_{(k,1^d)}(n)$ is exactly the multiplicity of the super-Schur module $S^{(k,1^d)}(\mathbb{C}^{1|1})$ in the super-coinvariant algebra $SR_n$.

6. For any hook of size $\geq 1$, the irreducible $GL(1|1)$ module $S^{(k,1^d)}(\mathbb{C}^{1|1})$ has dimension 2, spanned by a highest-weight vector and its lowering derivation. It is a canonical combinatorial theorem (e.g., Haglund–Rhoades–Shimozono) that the total dimension of $SR_n$ is exactly $\mathrm{OSP}(n)$, the number of Ordered Set Partitions of $[n]$.

7. **Combinatorial Interpretation:** The universal coefficient $c_{(k,1^d)}(n)$ exactly enumerates the number of Ordered Set Partitions of $[n]$ that act as highest-weight "sources" (elements annihilated by the fermionic derivation operator in the OSP combinatorial basis of $SR_n$) possessing exactly a polynomial degree of $k$ and an exterior degree of $d$.

*(Note on the $m = 1$ collapse: For $\mathrm{d} = 0$, the exterior degree is zero, which strictly filters the domain to Ordered Set Partitions possessing exactly $n$ blocks. An Ordered Set Partition of $[n]$ into exactly $n$ blocks is bijectively a permutation in $S_n$. Thus, for $\mathrm{d} = 0$, the combinatorial interpretation perfectly collapses to counting permutations by their major index $k$, yielding the exact Mahonian distribution $c_{(k)}(n)$ without violating cardinality bounds).* $\square$